

Digital Design Diploma

Assignment 4

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Question_1

1. Verilog code :

- module code:

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1 module ALSU (clk,rst,A,B,red_op_A,red_op_B,cin,serial_in,opcode,bypass_A,bypass_B,direction,out,leds);
2 parameter INPUT_PRIORITY = "A"; parameter FULL_ADDER = "ON";
3 input [2:0] A,B,opcode;
4 input clk,rst,cin,serial_in,red_op_A,red_op_B,bypass_A,bypass_B,direction;
5 output reg [5:0] out; output reg [15:0] leds;
6 wire [2:0] A_DFF,B_DFF,opcode_DFF;
7 wire cin_dff,serial_in_dff,red_op_A_dff,red_op_B_dff,bypass_A_dff,bypass_B_dff,direction_dff;
8 wire invalid_opcode = (opcode == 3'b110) || (opcode == 3'b111);
9 wire invalid_red_op = (red_op_A || red_op_B) && !(opcode == 3'b000 || opcode == 3'b001);
10 wire invalid_case = invalid_opcode || invalid_red_op;
11 D_FF #(3) dff1(clk,rst,A,A_DFF);
12 D_FF #(3) dff2(clk,rst,B,B_DFF);
13 D_FF #(3) dff3(clk,rst,opcode,opcode_DFF);
14 D_FF #(1) dff5(clk,rst,cin,cin_dff);
15 D_FF #(1) dff6(clk,rst,serial_in,serial_in_dff);
16 D_FF #(1) dff7(clk,rst,red_op_A,red_op_A_dff);
17 D_FF #(1) dff8(clk,rst,red_op_B,red_op_B_dff);
18 D_FF #(1) dff9(clk,rst,bypass_A,bypass_A_dff);
19 D_FF #(1) dff10(clk,rst,bypass_B,bypass_B_dff);
20 D_FF #(1) dff11(clk,rst,direction,direction_dff);
21 ///////////////////////////////////////////////////
22 always @(posedge clk or posedge rst) begin
23 if(rst) begin
24 out<=0;
25 end
26 else begin
27 if((bypass_A_dff==1)&&(bypass_B_dff==0))
28 begin
29 out=A_DFF;
30 end
31 else if ((bypass_B_dff==1)&&(bypass_A_dff==0))
32 begin
33 out=B_DFF;
34 end
35 else if (bypass_A_dff&&bypass_B_dff)
36 begin
37 if (INPUT_PRIORITY == "A")
38 out=A_DFF;
39 else
40 out=B_DFF;
41 end
42 else
43 begin
44 if((red_op_A_dff==1)&&(red_op_B_dff==0)) begin
45 case (opcode_DFF)
46 3'b000: out =A_DFF;
47 3'b001:out=A_DFF ;
48 3'b010: begin
49 if (FULL_ADDER == "ON")
50 out=A_DFF+B_DFF+cin_dff;
51 else
52 out=A_DFF+B_DFF;
53 end
54 3'b011:out=A_DFF*B_DFF;
55 3'b100: begin
```

```

56   if(direction_dff)
57     out={out[4:0],serial_in_dff};
58   else
59     out={serial_in_dff,out[5:1]};
60   end
61   3'b101: begin
62     if(direction_dff)
63       out={out [4:0],out[5]} ;
64     else
65       out={out [0],out[5:1]};
66     end
67     3'b110:out=6'b0;
68     3'b111:out=6'b0;
69   endcase
70   end
71   else if ((red_op_B_dff==1)&&(red_op_A_dff==0))begin
72     case (opcode_DFF)
73       3'b000: out =8B_DFF;
74       3'b001:out=~B_DFF ;
75       3'b010: begin
76         if (FULL_ADDER == "ON")
77           out=A_DFF+B_DFF+cin_dff;
78         else
79           out=A_DFF+B_DFF;
80         end
81         3'b011:out=A_DFF*B_DFF;
82       3'b100: begin
83         if(direction_dff)
84           out={out[4:0],serial_in_dff};
85         else
86           out={serial_in_dff,out[5:1]};
87         end
88       3'b101: begin
89         if(direction_dff)
90           out={out [4:0],out[5]} ;
91         else
92           out={out [0],out[5:1]};
93         end
94       3'b110:out=6'b0;
95       3'b111:out=6'b0;
96     endcase
97   end
98   else if (red_op_A_dff&&red_op_B_dff) begin
99     case (opcode_DFF)
100       3'b000: begin
101         if(INPUT_PRIORITY == "A")
102           out=8A_DFF;
103         else begin
104           out =8B_DFF;
105         end
106       end
107       3'b001: begin
108         if(INPUT_PRIORITY == "A")
109           out ={5'b0, ^A_DFF};

```

```

110     else begin
111       out ={5'b0, ^B_DFF};
112     end
113   end
114   3'b010: begin
115     if (FULL_ADDER == "ON")
116       out=A_DFF+B_DFF+cin_dff;
117     else
118       out=A_DFF+B_DFF;
119     end
120     3'b011:out=A_DFF*B_DFF;
121     3'b100: begin
122       if(direction_dff)
123         out={out[4:0],serial_in_dff};
124       else
125         out={serial_in_dff,out[5:1]};
126       end
127     3'b101:begin
128       if(direction_dff)
129         out={out [4:0],out[5]} ;
130       else
131         out={out [0],out[5:1]};
132       end
133     3'b110:out=6'b0;
134     3'b111:out=6'b0;
135   endcase
136   end
137   else begin
138     case (opcode_DFF)
139       3'b000: out =A_DFF&B_DFF;
140       3'b001:out=A_DFF*B_DFF;
141       3'b010: begin
142         if (FULL_ADDER == "ON")
143           out=A_DFF+B_DFF+cin_dff;
144         else
145           out=A_DFF+B_DFF;
146         end
147       3'b011:out=A_DFF*B_DFF;
148       3'b100:begin
149         if(direction_dff)
150           out={out[4:0],serial_in_dff};
151         else
152           out={serial_in_dff,out[5:1]};
153         end
154       3'b101:begin
155         if(direction_dff)
156           out={out [4:0],out[5]} ;
157         else
158           out={out [0],out[5:1]};
159         end
160       3'b110:out=6'b0;
161       3'b111:out=6'b0;
162     endcase
163   end
164   end

```

```

165     end
166   end
167   always @(posedge clk or posedge rst) begin
168     if (rst)
169       leds <= 16'b0; // Reset LEDs
170     else if (invalid_case)
171       leds <= ~leds; // Toggle all bits to make them blink
172     else
173       leds <= 16'b0; // Turn off LEDs for valid cases
174   end
175 endmodule

```

- test bench :

```

1 module ALSU tb();
2 parameter INPUT_PRIORITY = "A"; parameter FULL_ADDER = "ON";
3 reg [2:0] A,B,opcode;
4 reg clk,rst,cin,serial_in,red_op_A,red_op_B,bypass_A,bypass_B,direction;
5 wire [5:0] out_dut; wire[15:0] leds_dut;
6
7 ALSU alsu (clk,rst,A,B,red_op_A,red_op_B,cin,serial_in,opcode,bypass_A,bypass_B,direction,out_dut,leds_dut);
8
9 initial begin
10   clk=0;
11   forever
12     #1 clk=~clk;
13   end
14
15   integer i;
16   initial begin
17     rst=1;
18     A=0; B=0; opcode=0;
19     cin=0; serial_in=0; red_op_A=0; red_op_B=0; bypass_A=0; bypass_B=0; direction=0;
20     @(negedge clk);
21     if((out_dut!=0'b0)|| (leds_dut!=16'b0)) begin
22       $display ("error");
23       $stop;
24     end
25   end begin
26     $display ("rst=%b,A=%b,B=%b,opcode=%b,cin=%b,serial_in=%b,red_op_A=%b,red_op_B=%b,bypass_A=%b,bypass_B=%b,direction=%b,out_dut=%b,leds_dut=%b",rst,A,B,opcode,cin,serial_in,red_op_A,red_op_B,bypass_A,bypass_B,direction,out_dut,leds_dut);
27   end
28   rst=0; bypass_A=1; bypass_B=1;
29
30   for(i=0;i<10;i=i+1) begin
31     A=$random; B=$random; opcode=$urandom_range(0,5);
32     repeat (2) begin
33       @(negedge clk);
34     end
35     if(out_dut!=A) begin
36       $display("error");
37       $stop;
38     end
39     else
40       $display ("rst=%b,A=%b,B=%b,opcode=%b,cin=%b,serial_in=%b,red_op_A=%b,red_op_B=%b,bypass_A=%b,bypass_B=%b,direction=%b,out_dut=%b,leds_dut=%b",rst,A,B,opcode,cin,serial_in,red_op_A,red_op_B,bypass_A,bypass_B,direction,out_dut,leds_dut);
41     end
42     bypass_A=0; bypass_B=0; opcode=3'b0;
43     for(i=0;i<10;i=i+1) begin
44       A=$random; B=$random; red_op_A=$random; red_op_B=$random;
45       repeat (2) begin
46         @(negedge clk);
47       end
48       if(((red_op_A==1)&&(out_dut!=8A))||((red_op_B==1)&&(out_dut!=8B))) begin
49         $display("error");
50         $stop;
51       end
52       else if ((red_op_A==1)&&(red_op_B==1)&&(out_dut!=8A)) begin
53         $display("error");
54         $stop;
55       end
56       else if((red_op_A==0)&&(red_op_B==0)&&(out_dut!= {3'b0, A & B})) begin
57         $display("error");
58         $stop;
59       end
60       else begin
61         $display ("rst=%b,A=%b,B=%b,opcode=%b,cin=%b,serial_in=%b,red_op_A=%b,red_op_B=%b,bypass_A=%b,bypass_B=%b,direction=%b,out_dut=%b,leds_dut=%b",rst,A,B,opcode,cin,serial_in,red_op_A,red_op_B,bypass_A,bypass_B,direction,out_dut,leds_dut);
62       end
63       opcode=3'b001;
64       for (i=0;i<10;i=i+1) begin
65         A=$random; B=$random;
66         red_op_A=$random; red_op_B=$random;
67         repeat (2) begin
68           @(negedge clk);
69         end
70         if((red_op_A==1)&&(red_op_B==0)&&(out_dut!= {3'b0, "A"})) begin
71           $display("error");
72           $stop;
73         end
74         else if ((red_op_A==1)&&(red_op_B==1)&&(out_dut!= {3'b0, "B"})) begin
75           $display("error");
76           $stop;
77         end
78         else if (((red_op_A==1)&&(red_op_B==1)&&(out_dut!= {3'b0, "A"})) begin
79           $display("error");
80           $stop;
81         end
82         else if((red_op_A==0)&&(red_op_B==0)&&(out_dut!= {3'b0, A + B})) begin
83           $display("error");
84           $stop;
85         end
86       end
87       else begin
88         $display ("rst=%b,A=%b,B=%b,opcode=%b,cin=%b,serial_in=%b,red_op_A=%b,red_op_B=%b,bypass_A=%b,bypass_B=%b,direction=%b,out_dut=%b,leds_dut=%b",rst,A,B,opcode,cin,serial_in,red_op_A,red_op_B,bypass_A,bypass_B,direction,out_dut,leds_dut);
89       end
90       opcode=3'b010;
91       for (i=0;i<10;i=i+1) begin
92         A=$random; B=$random; cin=$random;
93         repeat (2) begin
94           @(negedge clk);
95         end
96         if(out_dut!=(A+B+cin)) begin
97           $display("error");
98           $stop;
99         end
100       end
101       else
102         $display ("rst=%b,A=%b,B=%b,opcode=%b,cin=%b,serial_in=%b,red_op_A=%b,red_op_B=%b,bypass_A=%b,bypass_B=%b,direction=%b,out_dut=%b,leds_dut=%b",rst,A,B,opcode,cin,serial_in,red_op_A,red_op_B,bypass_A,bypass_B,direction,out_dut,leds_dut);
103       end
104       opcode=3'b011;
105       for (i=0;i<10;i=i+1) begin
106         A=$random; B=$random;
107         repeat (2) begin

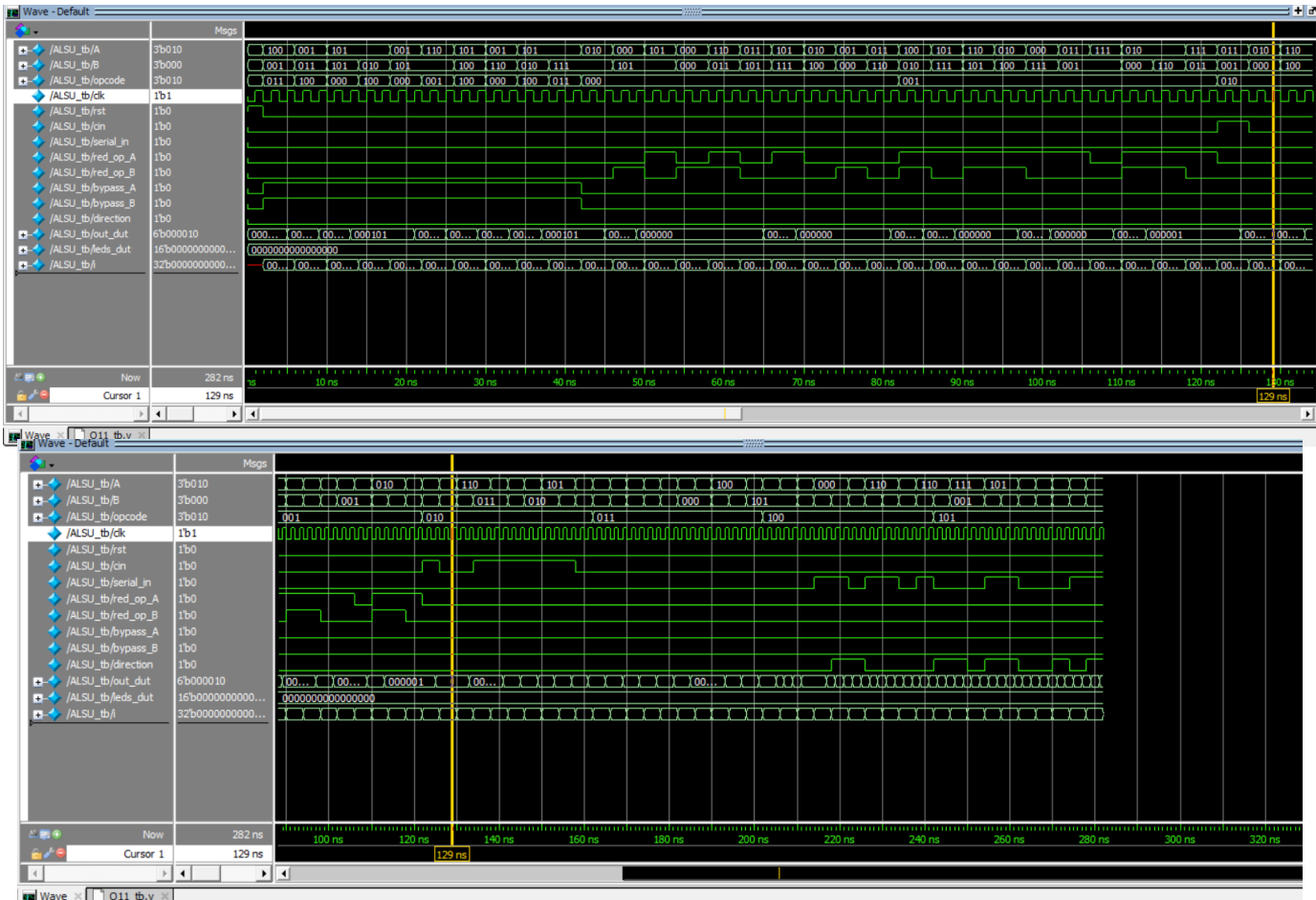
```

```

107 A=$random; B=$random;
108 repeat (2) begin
109   @(negedge clk);
110   end
111   if(out_dut!=A*B) begin
112     $display("error");
113   $stop;
114   end
115   else
116     $display ("rst=%b,A=%b,B=%b,opcode=%b,cin=%b,serial_in=%b,red_op_A=%b,red_op_B=%b,bypass_A=%b,bypass_B=%b,direction=%b,out_dut=%b,leds_dut=%b",rst,A,B,opcode,cin,serial_in,red_op_A,red_op_B,bypass_A,bypass_B,direction,
117 out_dut,leds_dut);
118   end
119   opcode=3'b100;
120   for (i=0;i<10;i=i+1) begin
121     A=$random; B=$random; serial_in=$random; direction=$random;
122     repeat (2) begin
123       @(negedge clk);
124       end
125       $display ("rst=%b,A=%b,B=%b,opcode=%b,cin=%b,serial_in=%b,red_op_A=%b,red_op_B=%b,bypass_A=%b,bypass_B=%b,direction=%b,out_dut=%b,leds_dut=%b",rst,A,B,opcode,cin,serial_in,red_op_A,red_op_B,bypass_A,bypass_B,direction,
126 out_dut,leds_dut);
127     opcode=3'b101;
128     for (i=0;i<10;i=i+1) begin
129       A=$random; B=$random; serial_in=$random; direction=$random;
130       repeat (2) begin
131         @(negedge clk);
132         end
133         $display ("rst=%b,A=%b,B=%b,opcode=%b,cin=%b,serial_in=%b,red_op_A=%b,red_op_B=%b,bypass_A=%b,bypass_B=%b,direction=%b,out_dut=%b,leds_dut=%b",rst,A,B,opcode,cin,serial_in,red_op_A,red_op_B,bypass_A,bypass_B,direction,
134 out_dut,leds_dut);
135       $stop;
136     end
137   endmodule

```

- **Waveform:**



- **Transcript:**

```
# rst=0,A=111,B=010,opcode=011,cin=0,serial_in=0,red_op_A=0,red_op_B=0,bypass_A=0,bypass_B=0,direction=0,out_dut=001110,leds_dut=0000000000000000
# rst=0,A=110,B=000,opcode=011,cin=0,serial_in=0,red_op_A=0,red_op_B=0,bypass_A=0,bypass_B=0,direction=0,out_dut=000000,leds_dut=0000000000000000
# rst=0,A=001,B=000,opcode=011,cin=0,serial_in=0,red_op_A=0,red_op_B=0,bypass_A=0,bypass_B=0,direction=0,out_dut=000000,leds_dut=0000000000000000
# rst=0,A=100,B=011,opcode=011,cin=0,serial_in=0,red_op_A=0,red_op_B=0,bypass_A=0,bypass_B=0,direction=0,out_dut=001100,leds_dut=0000000000000000
# rst=0,A=100,B=001,opcode=011,cin=0,serial_in=0,red_op_A=0,red_op_B=0,bypass_A=0,bypass_B=0,direction=0,out_dut=000100,leds_dut=0000000000000000
# rst=0,A=011,B=101,opcode=011,cin=0,serial_in=0,red_op_A=0,red_op_B=0,bypass_A=0,bypass_B=0,direction=0,out_dut=001111,leds_dut=0000000000000000
# rst=0,A=001,B=101,opcode=100,cin=0,serial_in=0,red_op_A=0,red_op_B=0,bypass_A=0,bypass_B=0,direction=0,out_dut=000111,leds_dut=0000000000000000
# rst=0,A=110,B=101,opcode=100,cin=0,serial_in=0,red_op_A=0,red_op_B=0,bypass_A=0,bypass_B=0,direction=0,out_dut=000001,leds_dut=0000000000000000
# rst=0,A=111,B=001,opcode=100,cin=0,serial_in=0,red_op_A=0,red_op_B=0,bypass_A=0,bypass_B=0,direction=0,out_dut=000000,leds_dut=0000000000000000
# rst=0,A=000,B=101,opcode=100,cin=0,serial_in=1,red_op_A=0,red_op_B=0,bypass_A=0,bypass_B=0,direction=0,out_dut=100000,leds_dut=0000000000000000
# rst=0,A=000,B=100,opcode=100,cin=0,serial_in=1,red_op_A=0,red_op_B=0,bypass_A=0,bypass_B=0,direction=1,out_dut=100001,leds_dut=0000000000000000
# rst=0,A=100,B=110,opcode=100,cin=0,serial_in=0,red_op_A=0,red_op_B=0,bypass_A=0,bypass_B=0,direction=1,out_dut=000110,leds_dut=0000000000000000
# rst=0,A=110,B=000,opcode=100,cin=0,serial_in=1,red_op_A=0,red_op_B=0,bypass_A=0,bypass_B=0,direction=0,out_dut=100110,leds_dut=0000000000000000
# rst=0,A=110,B=010,opcode=100,cin=0,serial_in=1,red_op_A=0,red_op_B=0,bypass_A=0,bypass_B=0,direction=0,out_dut=111001,leds_dut=0000000000000000
# rst=0,A=001,B=111,opcode=100,cin=0,serial_in=0,red_op_A=0,red_op_B=0,bypass_A=0,bypass_B=0,direction=0,out_dut=011110,leds_dut=0000000000000000
# rst=0,A=110,B=110,opcode=100,cin=0,serial_in=1,red_op_A=0,red_op_B=0,bypass_A=0,bypass_B=0,direction=0,out_dut=100111,leds_dut=0000000000000000
# rst=0,A=110,B=000,opcode=101,cin=0,serial_in=0,red_op_A=0,red_op_B=0,bypass_A=0,bypass_B=0,direction=1,out_dut=100111,leds_dut=0000000000000000
# rst=0,A=111,B=001,opcode=101,cin=0,serial_in=0,red_op_A=0,red_op_B=0,bypass_A=0,bypass_B=0,direction=1,out_dut=011110,leds_dut=0000000000000000
# rst=0,A=111,B=001,opcode=101,cin=0,serial_in=0,red_op_A=0,red_op_B=0,bypass_A=0,bypass_B=0,direction=0,out_dut=011110,leds_dut=0000000000000000
# rst=0,A=101,B=101,opcode=101,cin=0,serial_in=1,red_op_A=0,red_op_B=0,bypass_A=0,bypass_B=0,direction=1,out_dut=011110,leds_dut=0000000000000000
# rst=0,A=101,B=000,opcode=101,cin=0,serial_in=1,red_op_A=0,red_op_B=0,bypass_A=0,bypass_B=0,direction=1,out_dut=111001,leds_dut=0000000000000000
# rst=0,A=100,B=010,opcode=101,cin=0,serial_in=0,red_op_A=0,red_op_B=0,bypass_A=0,bypass_B=0,direction=0,out_dut=111001,leds_dut=0000000000000000
# rst=0,A=001,B=001,opcode=101,cin=0,serial_in=0,red_op_A=0,red_op_B=0,bypass_A=0,bypass_B=0,direction=0,out_dut=011110,leds_dut=0000000000000000
# rst=0,A=111,B=010,opcode=101,cin=0,serial_in=0,red_op_A=0,red_op_B=0,bypass_A=0,bypass_B=0,direction=1,out_dut=011110,leds_dut=0000000000000000
# rst=0,A=110,B=000,opcode=101,cin=0,serial_in=1,red_op_A=0,red_op_B=0,bypass_A=0,bypass_B=0,direction=0,out_dut=011110,leds_dut=0000000000000000
# rst=0,A=010,B=011,opcode=101,cin=0,serial_in=1,red_op_A=0,red_op_B=0,bypass_A=0,bypass_B=0,direction=1,out_dut=011110,leds_dut=0000000000000000
# ** Note: $stop      : D:\KareemWaseem\Questa sem\Assignment 4\Q11 tb.v(135)
```

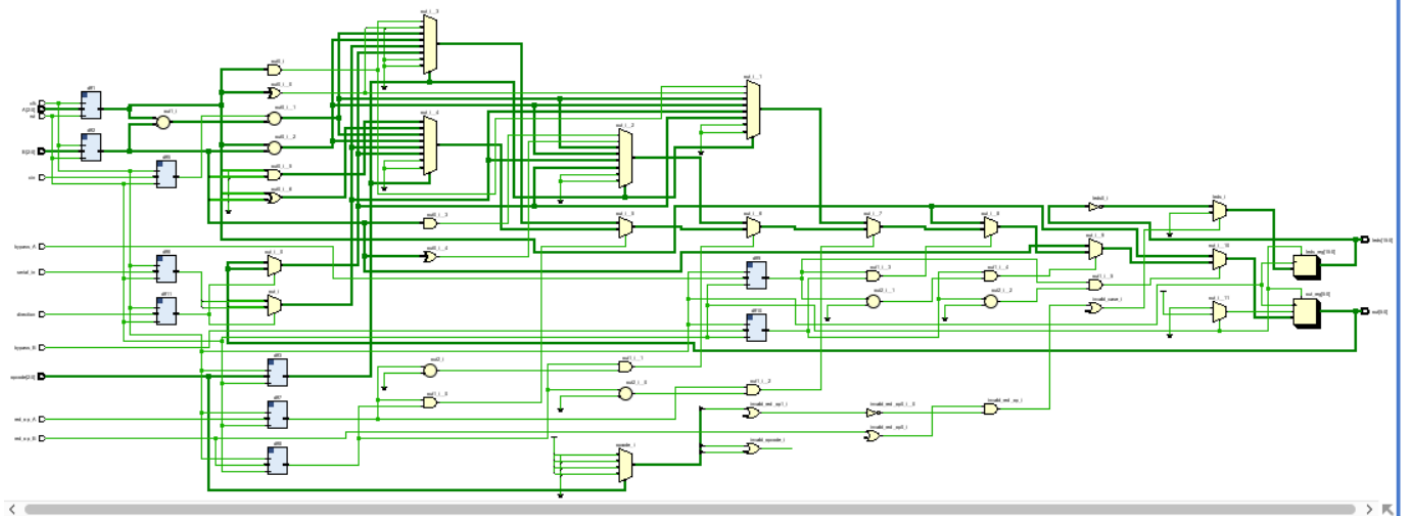
- **DO file:**

```
vlib work
vlog ALSU_DFF.v Q1_FINAL.v Q11_tb.v
vsim -voptargs=+acc work.ALSU_tb
add wave *
run -all
#quit -sim
```

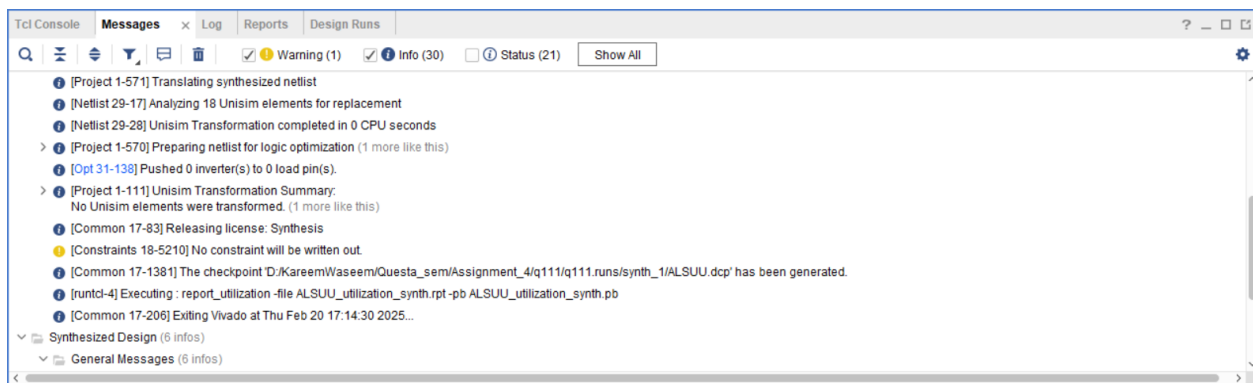
3. Synthesis tool:

- **Elaboration:**

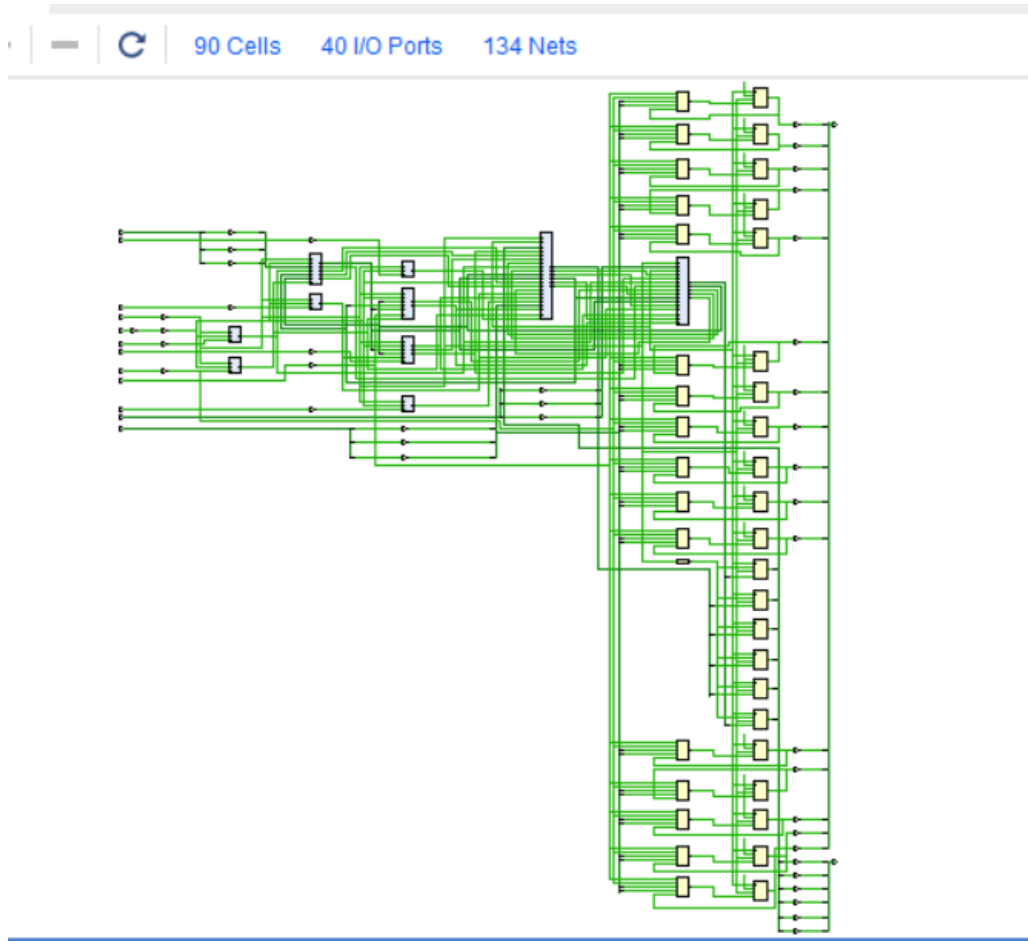




- **Synthesis :**
1. **Message :**



2. **Schematic:**



3. Report utilization:

Name	Slice LUTs (20800)	Slice Registers (41600)	Bonded IOB (106)	BUFGCTRL (32)	
ALSUU	49	38	40	1	
dff1 (D_FF)	8	3	0	0	
dff2 (D_FF_1)	7	3	0	0	
dff3 (D_FF_2)	15	3	0	0	
dff5 (D_FF__paramete...	0	1	0	0	
dff6 (D_FF__paramete...	0	1	0	0	
dff7 (D_FF__paramete...	1	1	0	0	
dff8 (D_FF__paramete...	0	1	0	0	
dff9 (D_FF__paramete...	0	1	0	0	

4. Timing report:

Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 6.491 ns	Worst Hold Slack (WHS): 0.139 ns	Worst Pulse Width Slack (WPWS): 4.500 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 22	Total Number of Endpoints: 22	Total Number of Endpoints: 39

All user specified timing constraints are met.

5. Debug core:

```
161 create_debug_core u_ila_0 ila
162 set_property ALL_PROBE_SAME_MU true [get_debug_cores u_ila_0]
163 set_property ALL_PROBE_SAME_MU_CNT 1 [get_debug_cores u_ila_0]
164 set_property C_ADV_TRIGGER false [get_debug_cores u_ila_0]
165 set_property C_DATA_DEPTH 1024 [get_debug_cores u_ila_0]
166 set_property C_EN_STRG_QUAL false [get_debug_cores u_ila_0]
167 set_property C_INPUT_PIPE_STAGES 0 [get_debug_cores u_ila_0]
168 set_property C_TRIGIN_EN false [get_debug_cores u_ila_0]
169 set_property C_TRIGOUT_EN false [get_debug_cores u_ila_0]
170 set_property port_width 1 [get_debug_ports u_ila_0/clock]
171 connect_debug_port u_ila_0/clock [get_nets [list clk_IBUF_BUF]]
172 set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe0]
173 set_property port_width 3 [get_debug_ports u_ila_0/probe0]
174 connect_debug_port u_ila_0/probe0 [get_nets [list {B_IBUF[0]} {B_IBUF[1]} {B_IBUF[2]}]]
175 create_debug_port u_ila_0 probe
176 set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe1]
177 set_property port_width 6 [get_debug_ports u_ila_0/probe1]
178 connect_debug_port u_ila_0/probe1 [get_nets [list {out_OBUF[0]} {out_OBUF[1]} {out_OBUF[2]} {out_OBUF[3]} {out_OBUF[4]} {out_OBUF[5]}]]
179 create_debug_port u_ila_0 probe
180 set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe2]
181 set_property port_width 3 [get_debug_ports u_ila_0/probe2]
182 connect_debug_port u_ila_0/probe2 [get_nets [list {A_IBUF[0]} {A_IBUF[1]} {A_IBUF[2]}]]
183 create_debug_port u_ila_0 probe
184 set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe3]
185 set_property port_width 16 [get_debug_ports u_ila_0/probe3]
186 connect_debug_port u_ila_0/probe3 [get_nets [list {leds_OBUF[0]} {leds_OBUF[1]} {leds_OBUF[2]} {leds_OBUF[3]} {leds_OBUF[4]} {leds_OBUF[5]} {leds_OBUF[6]} {leds_OBUF[7]}]]
187 create_debug_port u_ila_0 probe
188 set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe4]
189 set_property port_width 3 [get_debug_ports u_ila_0/probe4]
190 connect_debug_port u_ila_0/probe4 [get_nets [list {opcode_IBUF[0]} {opcode_IBUF[1]} {opcode_IBUF[2]}]]
191 create_debug_port u_ila_0 probe
192 set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe5]
193 set_property port_width 1 [get_debug_ports u_ila_0/probe5]
194 connect_debug_port u_ila_0/probe5 [get_nets [list bypass_A_IBUF]]
195 create_debug_port u_ila_0 probe
196 set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe6]
197 set_property port_width 1 [get_debug_ports u_ila_0/probe6]
198 connect_debug_port u_ila_0/probe6 [get_nets [list bypass_B_IBUF]]
199 create_debug_port u_ila_0 probe
200 set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe7]
201 set_property port_width 1 [get_debug_ports u_ila_0/probe7]
202 connect_debug_port u_ila_0/probe7 [get_nets [list cin_IBUF]]
203 create_debug_port u_ila_0 probe
204 set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe8]
205 set_property port_width 1 [get_debug_ports u_ila_0/probe8]
206 connect_debug_port u_ila_0/probe8 [get_nets [list clk_IBUF]]
207
```

- Implementation:

1. Messages:

Tcl Console Messages x Log Reports Design Runs Power DRC Methodology Timing

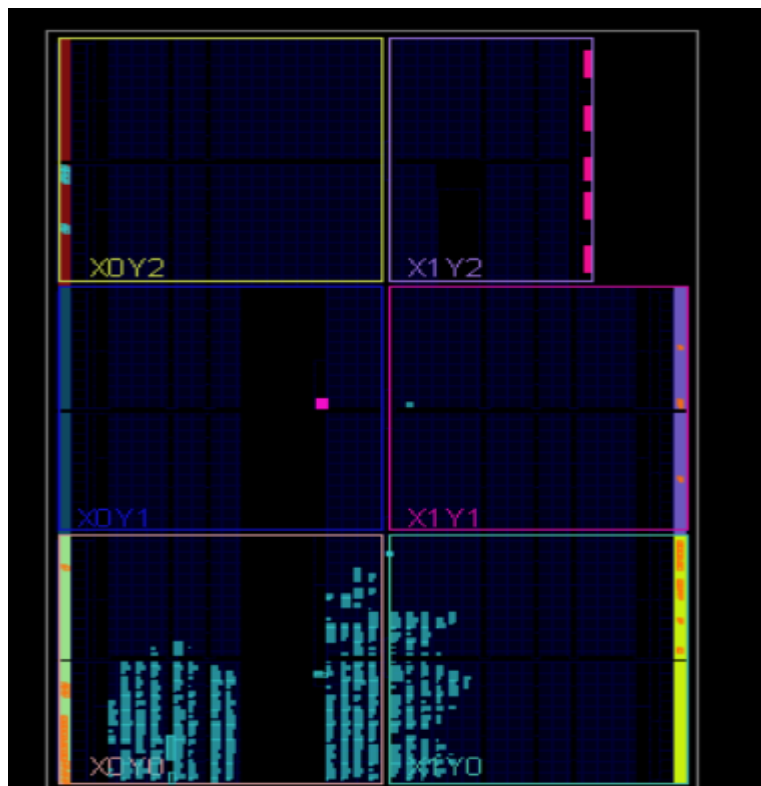
Warning (4) Info (248) Status (493) Show All

- Vivado Commands (3 infos)
- Synthesis (1 warning, 21 infos)
- Implementation (1 warning, 107 infos)
 - Design Initialization (11 infos)
 - [Netlist 29-17] Analyzing 18 Unisim elements for replacement
 - [Netlist 29-28] Unisim Transformation completed in 0 CPU seconds
 - [Project 1-479] Netlist was created with Vivado 2018.2
 - [Device 21-403] Loading part xc7a35ticpg236-1L
 - [Project 1-570] Preparing netlist for logic optimization

2. timing report:

Design Timing Summary			
Setup		Hold	Pulse Width
Worst Negative Slack (WNS): 2.502 ns		Worst Hold Slack (WHS): 0.061 ns	Worst Pulse Width Slack (WPWS): 3.750 ns
Total Negative Slack (TNS): 0.000 ns		Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0		Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 4884		Total Number of Endpoints: 4868	Total Number of Endpoints: 2864
All user specified timing constraints are met.			

3. Schematic:



4. Report utilization:

Name	Slice LUTs (20800)	Slice Registers (41600)	F7 Muxes (16300)	Slice (8150)	LUT as Logic (20800)	LUT as Memory (9600)	LUT Flip Flop Pairs (20800)	Block RAM Tile (50)	Bonded IOB (106)	BUFGCTRL (32)	BSCANE2 (4)
ALSUU	1697	2545	23	810	1520	177	1072	1.5	40	2	1
dbg_hub (dbg_hub)	475	727	0	237	451	24	311	0	0	1	1
dff1 (D_FF)	7	3	0	6	7	0	0	0	0	0	0
dff2 (D_FF_1)	7	3	0	5	7	0	0	0	0	0	0
dff3 (D_FF_2)	15	3	0	8	15	0	0	0	0	0	0
dff5 (D_FF__paramete...	0	1	0	1	0	0	0	0	0	0	0
dff6 (D_FF__escr...	0	1	0	1	0	0	0	0	0	0	0